Real-Time Adaptive Audio Interference Cancellation Using a Digitally Tunable Tow–Thomas Notch Filter

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***Abstract*—This paper presents a digitally controlled analog audio processing system that cancels out an interfering tone from a mixed audio signal in real time over the $2\,\mathrm{kHz}$ to $20\,\mathrm{kHz}$ band (human speech). A second-order band-reject filter with a notch response was created with a Tow–Thomas biquad filter. By replacing the integrator resistors in the circuit with digital potentiometers, control over the notch center frequency f_0 is realized with an Arduino Nano microcontroller. The interference signal is identified by a python script, taking in the FFT of the captured output from a Tektronix TDS2004B oscilloscope, and an Arduino Nano maps the detected peak frequency to a potentiometer code to retune the notch.**

***Index Terms*—Active filters, biquad, notch filter, adaptive filtering, digital potentiometer, mixed-signal, operational amplifiers.**

I. Introduction

As far as audio signal processing is concerned, the presence of interference signals greatly affects listening quality. In this case, implementation of a notch filter is a simple solution. However, in cases where the frequency of the interfering signals varies, a static notch cannot adapt to such changes, nor can it be designed to filter unexpected frequency components.

This ad-hoc project focuses on a simple analog embedded system consisting of an active filter architecture along with a digital control element. For the analog component, a symmetrical notch filter is obtained through a Tow–Thomas biquad filter structure using the integrator resistor value as the center frequency. Based on what was stated by Franco [1], the value of the center frequency can be changed by replacing the resistors in the integrators with digitally controlled potentiometers. A microcontroller is used to detect the frequency of the interference signal using FFT on the spectrum input and to adjust the value of the potentiometers to tune the notch.

II. System Architecture

Fig. 1 illustrates the experimental setup. The laptop is the source of the test input signal (music mixed with an interference tone). The Tektronix TDS2004B oscilloscope records the input signal, and sends the FFT to the laptop, which runs an algorithm to detect and forward the interference frequency to an Arduino Nano via a serial interface. Next, the Arduino adjusts the digital potentiometer values to retune the notch filter.

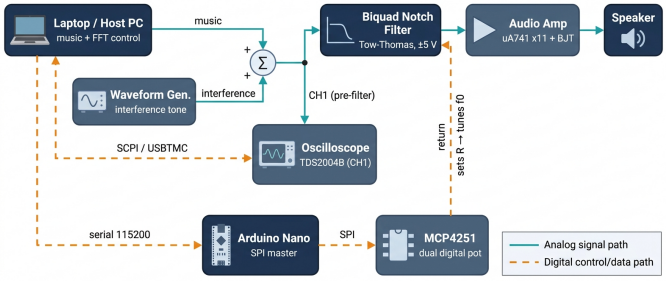


Fig. 1. System block diagram of the adaptive interference canceller.

III. Analog Notch Filter Design

A. Topology

The notch filter is based on the biquad structure of Franco [1, Fig. 3.37]: a notch is created using a two-integrator feedback loop plus an output summing block. As shown in Fig. 2, it uses four op amps, provided by the same OP484. OA_1 is a damped integrator that provides the band-pass term $-H_{\mathrm{BP}}V_i$. OA_2 is a lossless integrator to provide the low-pass term $H_{\mathrm{LP}}V_i$. OA_3 is a unity inverter, closing the two-integrator loop. Finally, OA_4 is the output summing amplifier.

The output signal is computed as

$$V_N = -\left[\frac{R_5}{R_2}\left(V_i - V_{\mathrm{BP}}\right) \pm \frac{R_5}{R_4}V_{\mathrm{LP}}\right], \tag{1}$$

in Franco's notation, with a plus sign and R_4 indicating a low-pass notch [1, eq. (3.85c)]. For the symmetric case where $R_4 \to \infty$ ($R_4 = \infty$), the term with V_{LP} disappears, simplifying the expression to subtracting V_{BP} from V_i.

B. Transfer Function

In general, from [1, eq. (3.85a)],

$$\frac{V_N}{V_i} = -\frac{R_5\,\omega_z^2}{R_2\,\omega_0^2}\cdot\frac{1-(\omega/\omega_z)^2}{1-(\omega/\omega_0)^2+j(\omega/\omega_0)/Q}, \tag{2}$$

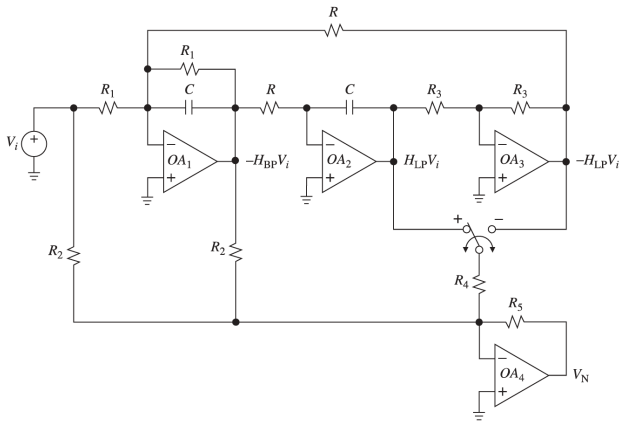


Fig. 2. Implemented Tow–Thomas notch filter. The four OP484 amplifiers OA_1–OA_4 map onto the two-integrator loop and output summer. The symmetric notch is realized by leaving R_4 open ($R_4 \to \infty$), which drops the V_{LP} term in (1).

with the notching frequency given by $\omega = \omega_z$. If $R_4 \to \infty$ (omitting the second term in the transfer function), then from [1, eq. (3.86)],

$$\omega_z = \omega_0, \qquad H_{0\mathrm{N}} = -\frac{R_5}{R_2}, \tag{3}$$

yielding the symmetric notch filter. It has an equal notch depth with passband gain $|H_{0\mathrm{N}}| = R_5/R_2$ (0 dB for $R_5 = R_2$).

C. Center Frequency and Q

If the integrator resistors are equal (R) and capacitors (C) are equal, then from [1, eq. (3.85b)],

$$\omega_0 = \frac{1}{RC} \;\Rightarrow\; f_0 = \frac{1}{2\pi RC}, \qquad Q = \frac{R_1}{R}, \tag{4}$$

with R_1 being the damping resistance (around OA_1). Under these circumstances, the -3 dB bandwidth is

$$\mathrm{BW} = \frac{f_0}{Q} = \frac{1}{2\pi R_1 C}, \tag{5}$$

(independent of R). Thus, tuning of f_0 via R keeps BW constant.

D. Digital Potentiometer as the Tuning Element

Each R represents a single section of the 8-bit ($N = 256$ tap) MCP4251 dual potentiometer, where the wiper divides the total resistance. Effective resistance as a function of D is expressed as

$$R(D) = R_{\mathrm{step}}\, D + R_W, \qquad D \in \{0, 1, \ldots, 256\}, \tag{6}$$

where the values of $R_{\mathrm{step}} \approx 35.8\,\Omega/\mathrm{tap}$ and wiper intercept $R_W \approx 184\,\Omega$ are specific to each section. When substituting $R(D)$ in the f_0 formula above, we obtain the commanded frequency

$$f_0(D) = \frac{1}{2\pi C\left(R_{\mathrm{step}} D + R_W\right)}, \tag{7}$$

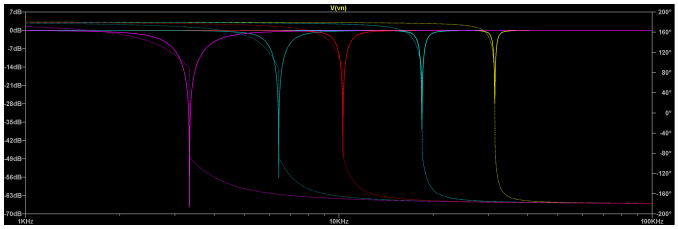


Fig. 3. Simulated notch magnitude response versus frequency for several potentiometer tap codes.

from which D can be found in terms of f_0 as follows:

$$D = \frac{1}{R_{\mathrm{step}}}\left(\frac{1}{2\pi C f_0} - R_W\right). \tag{8}$$

The inverse function is essential to the tuning algorithm.

E. Device Selection and Operating Conditions

All four amplifiers, OA_1–OA_4, are provided by the OP484 quad rail-to-rail amplifier operating on ± 5 V supplies, with one op amp occupying each of the nodes illustrated in Fig. 2. opAmps can be considered ideal for the purposes of circuit analysis. The value of C is chosen as 10 nF for both capacitors, while all the fixed resistors have the same value: $R_1 = R_2 = R_3 = R_5 = 10\,\mathrm{k}\Omega$. When $R_5 = R_2$, the notch passband has a gain of unity (0 dB), while matched resistors R_3 create the unity-gain inverting amplifier OA_3. Two integrator resistors, R, are created using the wipers of a single MCP4251 dual SPI digital potentiometer. The MCP4251 operates on the single 5 V/GND supply, independent of the ± 5 V supply.

IV. Adaptive Tuning Loop

A. Interference Detection

The input signal is multiplied with the Hann window function, and its Fourier transform is calculated using FFT. Peak detection is limited to the frequency band 2 kHz to 20 kHz, which excludes any musical signals. In order for a signal to be detected as a potential interferer, its magnitude must exceed a preset threshold, and it must consistently show up across 5 consecutive FFT frames.

B. Adjusting the Notch

The identified interference frequency is sent over SPI to the Arduino Nano, which converts it to a tap code to update the potentiometer wipers such that the notch sits atop the interferer.

V. Simulation Results

Notch filtering was simulated in LTspice using OP37 as a proxy for OP484. An `.ac lin` sweep with very fine resolution was performed to capture the narrow null. Fig. 3 shows the transfer function of the simulated notches for various tap codes.

TABLE I
COMMANDED VS. MEASURED NOTCH FREQUENCY

Tap code D	f_0 commanded (KHz)	f_0 measured (KHz)
217	2	2.10
84	5	5.02
39	10	11.23
17	20	20.73

TABLE II
PREDICTED VS. MEASURED PERFORMANCE

Metric	Simulated	Measured
Notch depth (dB)	(infinite)	-10 to -30
Q at 2 kHz	1.26	1.26
Q at 20 kHz	12.6	12.6

VI. EXPERIMENTAL RESULTS

A. Measured Frequency Response

The frequency response of the circuit correctly tuned to within approximately 1KHz of the target frequency. Small inconsistencies in the notch frequency are acceptable, as low Q factors allow neighboring frequencies to be significantly attenuated.

B. Frequency Calibration

Inaccuracies in frequency tuning from the potentiometer were addressed by sweeping across the frequency range and applying an inverse correction to the tuning algorithm. Table I summarizes the calibration.

C. Simulation vs. Measurement

Table II compares predicted and measured performance.

VII. DISCUSSION: TRADEOFFS AND SENSITIVITY

a) Q–f_0 coupling: Because the bandwidth is dependent on R_1 and C, the quality factor is also affected by f_0. For $R_1 = 10\,\mathrm{k\Omega}$ and $C = 10\,\mathrm{nF}$, the absolute bandwidth is $\mathrm{BW} = 1/(2\pi R_1 C) = 1.59\,\mathrm{kHz}$, so $Q = R_1/R = f_0/\mathrm{BW} = 1.3$ at $2\,\mathrm{kHz}$ and 12.6 at $20\,\mathrm{kHz}$, i.e., over a factor of 10 range. So, a given pair of resistance values constrains Q, which leads to widening of the notch at low frequencies and sharpening as f_0 increases toward $20\,\mathrm{kHz}$.

b) Detection limits: Effective cancellation of a stationary interference tone is a dance between two factors. On one hand, the notch should be as sharp as possible (high Q), such that neighboring frequencies are not noticeably attenuated. On the other hand, if the quality factor is too high any inconsistency in the tuned center frequency of the notch can cause the system to completely miss the target interference frequency. Further designs should explore novel solutions to this issue.

c) Interference Detection Speed: The TDS2004B has limited speed at which it can share data with the laptop. This creates a system bottleneck, drastically reducing the speed at which interference signals are detected. Future designs should explore alternatives, such as running all analysis and processing on a single device.

VIII. CONCLUSION AND FUTURE WORK

A digital Tow–Thomas filter with FFT adaptive control loop was designed, simulated, assembled and tested. Although the final design behaved as intended, it was slower than ideal, and interference frequencies were only partially attenuated.

Future work includes system optimizations such as combining all compute onto a single portable controller, and also implementing additional circuitry to improve interference cancellation. Finally, designing and printing a custom PCB will allow for the system to be used across many applications.

ACKNOWLEDGMENT

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REFERENCES

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